

MULTIMEDIA TECHNOLOGY

Unit-1: Introduction to Multimedia

Comprehensive Study Notes



1. Introduction to Multimedia

Definition

Multimedia is the combination of different content forms such as text, audio, images, animations, video, and interactive content to present information in a dynamic and engaging manner.

Key Concept: The word "Multimedia" comes from two words:

- **Multi** = Multiple
- **Media** = Medium of communication

Characteristics of Multimedia

1. **Integration** of multiple media types
2. **Digital representation**
3. **Interactive** capabilities
4. **User-controlled** navigation
5. **Non-linear** access to information

2. Needs and Areas of Use

Why Multimedia is Needed

Need	Description
Enhanced Communication	Combines visual and audio elements for better understanding
Effective Learning	Appeals to multiple senses, improves retention
Engaging Content	Captures attention better than static content
Interactive Experience	Allows user participation and exploration
Global Reach	Digital format enables easy distribution worldwide
Cost-Effective	Reduces printing and distribution costs
Dynamic Updates	Easy to update and modify content

Areas of Use

Sector	Applications
Education	E-learning, virtual classrooms, educational games, simulations
Entertainment	Movies, games, animation, virtual reality experiences
Business	Presentations, training modules, product demonstrations
Healthcare	Medical imaging, patient education, surgical simulations
Advertising	Digital billboards, interactive ads, product catalogs
Publishing	E-books, digital magazines, online newspapers

Government	Public information systems, citizen services
Tourism	Virtual tours, interactive maps, travel guides
Engineering	CAD visualization, simulation, prototyping
Research	Data visualization, scientific simulations

3. Development Platforms for Multimedia

A. Authoring Tools

- **Adobe Director** - For interactive multimedia applications
- **Adobe Flash** (Discontinued) - For animations and interactive content
- **Authorware** - For educational applications
- **ToolBook** - For CBT (Computer-Based Training)

B. Presentation Software

- **Microsoft PowerPoint** - Simple presentations
- **Apple Keynote** - Mac presentations
- **Prezi** - Zoomable presentations

C. Web-Based Platforms

- **HTML5 + CSS + JavaScript** - Web multimedia
- **WordPress** - Content management
- **Joomla** - CMS with multimedia support

D. Graphics & Design

- **Adobe Photoshop** - Image editing
- **Adobe Illustrator** - Vector graphics
- **CorelDRAW** - Graphic design

E. Animation & 3D

- **Adobe Animate** - 2D animation
- **Blender** - 3D modeling and animation
- **Maya** - Professional 3D animation
- **3ds Max** - 3D modeling

F. Audio Tools

- **Adobe Audition** - Audio editing
- **Audacity** - Free audio editing
- **FL Studio** - Music production

G. Video Tools

- **Adobe Premiere Pro** - Video editing
- **Final Cut Pro** - Mac video editing
- **DaVinci Resolve** - Color grading and editing

4. Elements of Multimedia

MULTIMEDIA ELEMENTS					
TEXT	IMAGES	SOUND	VIDEO	ANIMATION	
Plain	Vector	Mono	Analog	2D	
Formatted	Raster	Stereo	Digital	3D	
RTF	Bitmap	MP3	HDTV	Morphing	
HTML	JPEG	WAV	MPEG	Tweening	

Element	Description	Examples
Text	Written information displayed	Articles, titles, subtitles
Images	Visual representations	Photos, illustrations, diagrams
Sound	Audio content	Background music, narration
Animation	Moving graphics	Cartoons, transitions, effects
Video	Moving images with sound	Clips, tutorials, movies
Interactive	User-responsive elements	Buttons, links, forms

Text

Fonts: Typeface design (e.g., Arial, Times New Roman, Courier). Fonts can be serif (with small lines at ends) or sans-serif (without).

Font Size: Measured in points (pt) - 1pt = 1/72 inch. Common sizes: 10pt (body), 12pt (large body), 18pt+ (headings).

Font Color: RGB or hex values; used for emphasis, hierarchy, and aesthetic appeal.

Graphics

Feature	Bitmap (Raster)	Vector
Composition	Pixels (dots)	Mathematical paths
Quality when scaled	Loses quality (pixelates)	No loss of quality
File size	Large	Small
Formats	BMP, JPEG, PNG, GIF	SVG, EPS, AI, CDR
Best for	Photographs, complex images	Logos, illustrations, diagrams

Audio

Analog to Digital conversion (ADC): Sampling → Quantization → Encoding.

Sampling Rate: Number of samples per second (e.g., 44.1 kHz for CD quality).

Audio Formats:

Format	Type	Features
WAV	Uncompressed	High quality, large size
MP3	Compressed (lossy)	Small size, good quality
WMA	Compressed	Windows Media Audio
AAC	Compressed	Advanced Audio Coding
OGG	Open-source compressed	Free format

Video

Feature	Analog Video	Digital Video
Signal	Continuous electrical signal	Discrete binary data
Quality	Degrades with copying	No degradation
Storage	Magnetic tape	Hard drive, SSD, cloud
Examples	VHS, Betamax	DVD, Blu-ray, streaming

Video Formats & Codecs:

Format	Codec	Use
MP4	H.264 / H.265	Web, mobile, general
AVI	Various	Windows compatibility
MOV	ProRes, H.264	Apple ecosystem
WMV	Windows Media Video	Windows media
WebM	VP8 / VP9	Web (open-source)

Animation

- **2D Animation:** Flat, two-dimensional movement (e.g., cartoons, Flash animations)
- **3D Animation:** Three-dimensional objects with depth, shading, and realistic movement
- **Cel Animation:** Hand-drawn, frame-by-frame traditional animation
- **Path Animation:** Object moves along a predefined path/curve
- **Morphing:** One image smoothly transforms into another
- **Tweening:** Computer generates intermediate frames between keyframes

5. Making Simple Multimedia with PowerPoint

PowerPoint serves as a simple yet effective multimedia authoring tool supporting text, images, animation, video, sound, and hyperlinks.

Steps to Create Multimedia Presentation

1. **Plan** - Define purpose and audience
2. **Structure** - Create slide outline
3. **Add Text** - Include titles, bullet points
4. **Insert Images** - Add relevant visuals
5. **Add Animation** - Use transitions and effects
6. **Include Audio** - Add narration or music
7. **Insert Video** - Embed video clips
8. **Add Interactivity** - Insert hyperlinks and action buttons
9. **Review** - Check and rehearse
10. **Present** - Deliver the presentation

6. Concepts of Plain and Formatted Text

Plain Text vs Formatted Text

Feature	Plain Text	Formatted Text
Font Style	Fixed (monospace)	Variable
Font Size	Fixed	Variable
Color	Single	Multiple
Bold/Italic	Not supported	Supported
Paragraphs	Line breaks only	Alignment, spacing
Images	Not supported	Supported
Hyperlinks	Text only	Clickable
File Extension	.txt	.doc, .rtf, .html

7. RTF & HTML Texts

RTF (Rich Text Format)

- Developed by Microsoft
- Cross-platform compatibility
- Stores text with formatting information
- Uses control words and symbols in curly braces
- Intermediate format between plain text and word docs
- File extension: .rtf

Example RTF Code:

```
{\rtf1\ansi\deff0 {\fonttbl {\f0 Times New Roman;}}  
  
 \f0\fs24 This is \b bold \b0 and \i italic \i0 text.}
```

HTML (Hypertext Markup Language)

- Web page markup language
- Uses tags to format content
- Supports hyperlinks
- Can embed multimedia elements
- Rendered by web browsers
- File extension: .html, .htm

Example HTML Code:

```
<html>
```

```
<head><title>My Page</title></head>
```

```
<body>
```

```
<h1>Welcome</h1>
```

```
<p>This is <b>bold</b> and <i>italic</i> text.</p>
```

```
<a href="page2.html">Link</a>
```

```
</body>
```

```
</html>
```

RTF vs HTML Comparison

Feature	RTF	HTML
Purpose	Document exchange	Web content
Platform	Desktop applications	Web browsers
Hyperlinks	Limited	Native support
Multimedia	Limited	Full support
Scripting	Not supported	JavaScript support
CSS Support	No	Yes

8. Object Linking and Embedding (OLE)

OLE is a Microsoft technology for sharing objects between applications. Objects can be documents, images, or media.

Linking vs Embedding

Feature	Linking	Embedding
Storage	Source file only	Destination file
File Size	Smaller	Larger
Updates	Automatic when source changes	Manual, if updated
Independence	Dependent on source	Independent
Editing	Using source application	Using destination app
Portability	Need source file	Self-contained
Space	Saves space	Uses more space
Best For	Frequently updated content	Static content

Questions

Short Questions (2 marks)

1. Define multimedia.
.....
2. List the characteristics of multimedia.
.....
3. What is the difference between bitmap and vector graphics?
.....
4. Define OLE.
.....
5. What is RTF?
.....
6. What is sampling rate in audio?
.....
7. Differentiate between plain text and formatted text.
.....
8. Name any two audio formats.
.....
9. What is morphing?
.....
10. What is the difference between analog and digital video?
.....

Medium Questions (5 marks)

1. Explain the areas of use of multimedia in detail.
.....
2. Differentiate between linking and embedding in OLE.
.....
3. Explain the elements of multimedia with examples.
.....
4. Compare RTF and HTML text.
.....
5. Explain analog to digital conversion of audio.
.....

6. What are the different types of animation?

Long Questions (12 marks)

1. Explain multimedia in detail, including its definition, characteristics, and applications in various fields.

2. What are the elements of multimedia? Explain each element in detail with examples.

3. Explain OLE technology in detail. Compare linking and embedding with advantages and disadvantages.

4. Discuss various development platforms and authoring tools used for multimedia.

End of Unit I Notes